PUBLIC DATASET • NO WET LAB

lncRNA glow-up check.

GSE159185 ka deposited data dekh rahe hain: p53 overexpression ke baad lncRNA ka behavior kaisa hai.

TL;DR

4 stars jo har p53 condition mein upar gaye: SNHG4, DLG2AS, lincRNA-p21, lincRNA-VLDLR.

SNHG4 WT p53 mein 594.01x — sabse bada number, lekin control ke kareeb signal ho to aise numbers zyada ajeeb lagte hain.

Data = 1 value per target per condition. Replicates nahi, statistics nahi, sirf descriptive vibes.

33	6	4	594x
targets with values	p53 conditions	highlighted lncRNAs	top WT fold change

01 • THE SETUP

Dataset aur scope

Uploaded matrices mein 95 rows (target/reference entries) aur seven condition columns hain. Non-normalized file mein Ct-like values hain; "Undetermined" entries missing/non-detected hain aur numeric value mein convert nahi ki gayi. Fold-change file mein pcDNA6A-Flag control ko 1.00 set kiya gaya hai.

Conditions: pcDNA6A-Flag control, Wtp53, aur five mutant p53: R175H, L194F, R273H, R280K, E285K. Normalization reference 18S rRNA hai, jaisa deposited metadata batata hai.

Reality check

Ek value per target per condition hai. Biological variance estimate nahi ho sakti, isliye yahan koi p-value, confidence interval ya "significant" claim nahi hai. Sirf descriptive comparisons.

02 • UP OR DOWN?

Direction ka scene

Control = 1.00. Upar (>1) ya neeche (<1) classify kiya gaya. Yeh counts descriptive hain, tested effects nahi.

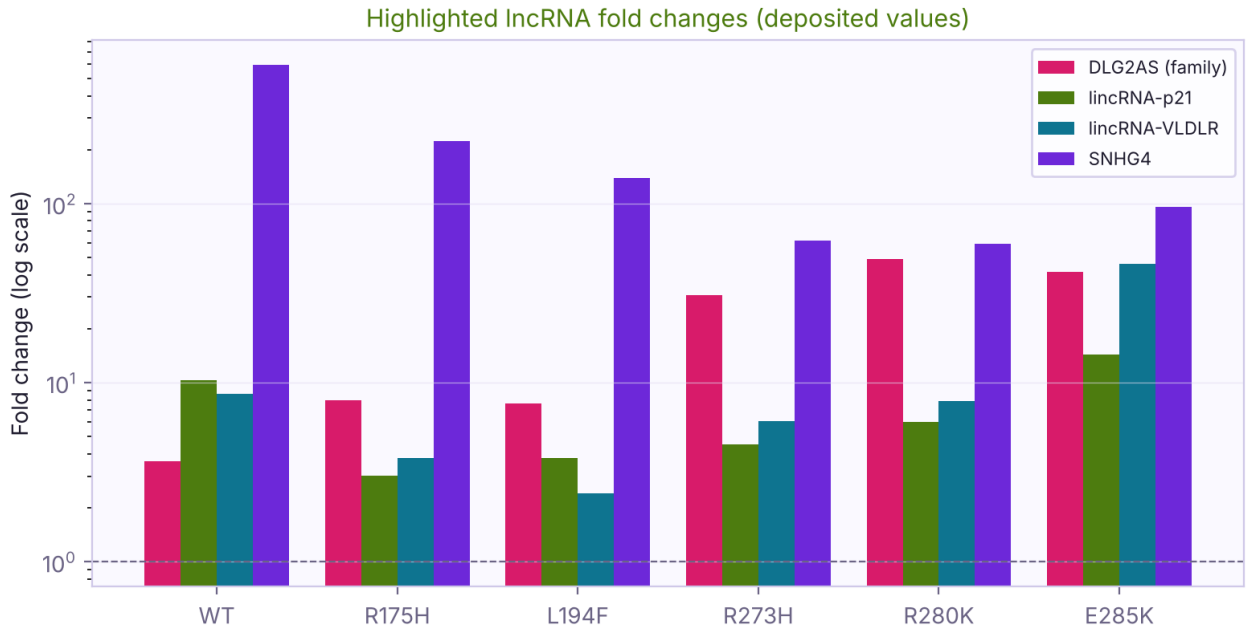
Condition	Targets	Above 1.0	Below 1.0
Wtp53	33	16	17
Mutp53 (R175H)	33	12	21
Mutp53 (L194F)	33	12	21
Mutp53 (R273H)	33	17	16
Mutp53 (R280K)	33	22	11
Mutp53 (E285K)	33	27	6

Vibe: E285K sabse "up" wali condition hai (27 up, sirf 6 down).

Four highlighted lncRNAs

Yeh chaaron targets sabhi six p53 conditions mein 1.0 se upar hain. Values uploaded fold-change file se li gayi hain.

Target	WT	R175H	L194F	R273H	R280K	E285K
DLG2AS (family)	3.64	7.99	7.66	30.80	49.19	41.37
lincRNA-p21	10.34	3.01	3.77	4.52	6.01	14.38
lincRNA-VLDLR	8.68	3.80	2.40	6.07	7.86	46.19
SNHG4	594.01	224.10	139.25	62.06	59.76	95.62



Quick highlights

- SNHG4 ka WT fold change sabse bada: **594.01**.
- DLG2AS (family) peak: **49.19** in R280K.
- lincRNA-VLDLR peak: **46.19** in E285K.
- lincRNA-p21 peak: **14.38** in E285K.

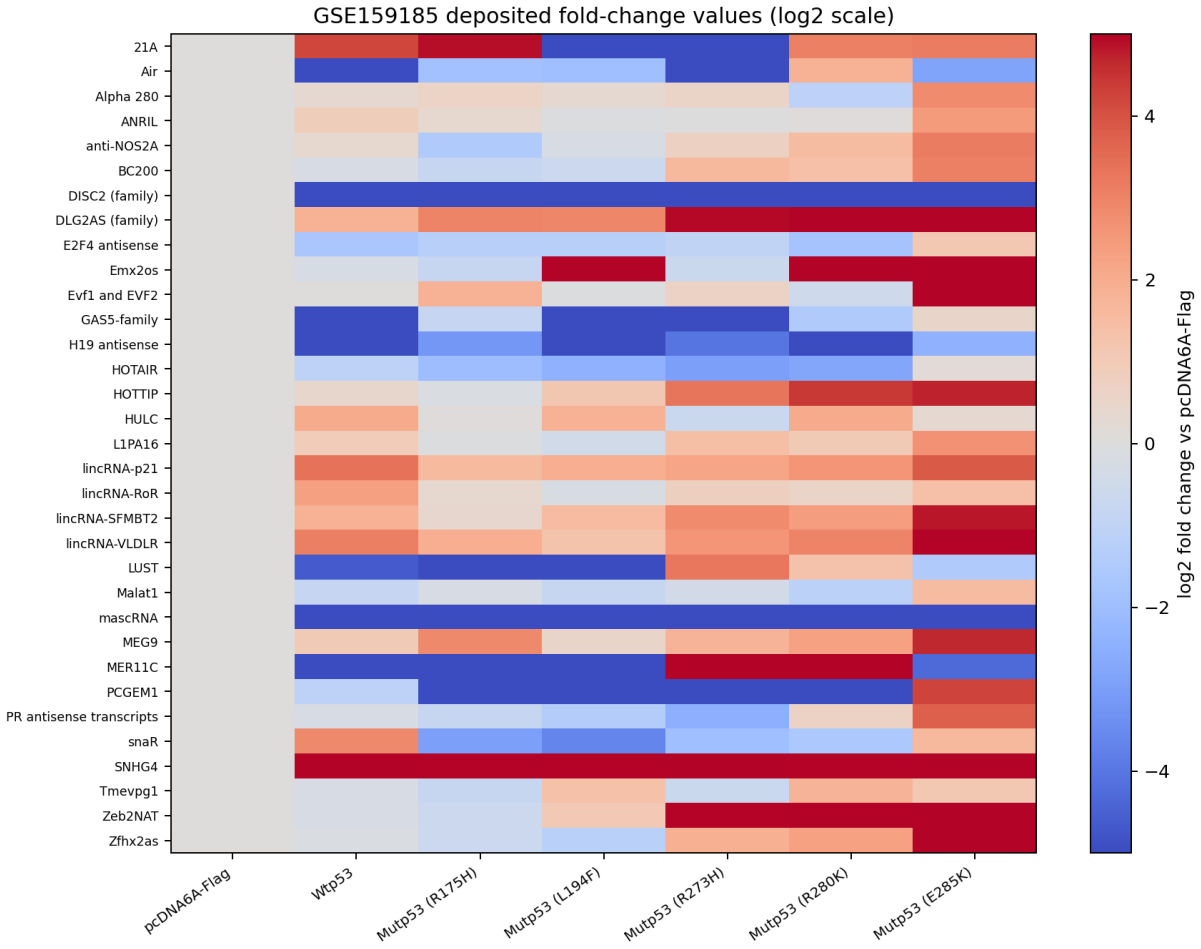
Broader pattern

Single-condition ke sabse strong fold changes:

Target	Condition	Fold change
Emx2os	Mutp53 (E285K)	1,007.95
SNHG4	Wtp53	594.01
Zeb2NAT	Mutp53 (R273H)	153.06

Caution label

Yeh condition-specific observations hain. Inhe replicated ya statistically significant effects samajhna galat hoga. Heatmap ke liye log2 scale use hua hai, aur color range ±5 par cap hai.



Original heatmap from the uploaded report (log2 scale, capped at ±5).

05 • VIBE CHECK

Interpretation

p53 overexpression conditions ke saath lncRNA fold-change patterns mukhtalif hain. Highlighted lncRNAs (DLG2AS, lincRNA-p21, lincRNA-VLDLR, SNHG4) sabhi six p53 conditions mein control se upar hain. Magnitude mutant ke hisaab se badalta hai, jo variant-dependent pattern ka ishara hai aur aage careful follow-up maangta hai.

Bohot bade fold changes (jaise SNHG4 in WT aur Emx2os in E285K) ko carefully treat karna hai. Jab control signal assay detection limit ke qareeb ho to ratios bohot bade ho sakte hain. Sirf yeh files replicate-level uncertainty nahi batati.

06 • FINE PRINT

Methodological limitations

- Condition-level values hain, replicate-level nahi. Koi inferential statistics nahi ki gayi.
- "Undetermined" entries missing rakhi gayi hain; zero se impute nahi ki gayi.
- Fold changes deposited fold-change file se liye gaye, Ct se recalculate nahi kiye gaye.
- Yeh secondary computational analysis hai, naya wet-lab experiment nahi.
- Biological conclusions ke liye original publication aur full GEO metadata check karein.

07 • THE TAKEAWAY

Conclusion

Supplied GSE159185 matrices mein 33 targets ki reported fold-change values hain. SNHG4, DLG2AS, lincRNA-p21 aur lincRNA-VLDLR mein prominent increases dikhte hain, aur Emx2os aur Zeb2NAT mein condition-specific high fold changes hain. Replicate data na hone ki wajah se significance testing possible nahi.

Sources

Data source: NCBI Gene Expression Omnibus, accession GSE159185. Input files: GSE159185_non-normalized_data.txt.gz and GSE159185_fold-change_data.txt.gz.

Reference context: He et al. (2021), Molecular Therapy – Nucleic Acids. Final submission mein official GEO aur article records cite karein.